MINI PROJECTS 1–4

A combined presentation

Computer hardware • SQL databases • Networking • Cybersecurity awareness

Presentation overview

Project 1 — Computer fundamentals

- Workstation builds for four job roles
- Hardware selection, cost and budget
- SQL database design and reporting

Projects 2–4 — Software & security

- Project 2: Pennine Pizza Works ordering program
- Project 3: LAN, IP addressing and connectivity
- Project 4: device safety, authentication, public Wi-Fi and phishing

Project 1 | Northwest Architecture Studio Ltd.

- Goal: recommend suitable computer workstations for different staff roles.
- Roles covered: Administrative Assistant, Project Manager, Architect/CAD Designer and 3D Visualisation Specialist.
- Match hardware capability to the work performed while controlling the hardware refresh cost.
- SQL tables and queries support the workstation inventory, roles, recommendations and cost summaries.

Project 1 | Match the workstation to the role

Office and coordination roles

- Administrative Assistant: documents, email, records and video calls.
- Project Manager: spreadsheets, reports, schedules, dashboards and meetings.
- Priorities: dependable CPU, sufficient RAM, fast SSD, clear display and reliable networking.

Specialist design roles

- Architect/CAD Designer: CAD and technical design workloads.
- 3D Visualisation Specialist: demanding graphics and visualisation tasks.
- Priorities: stronger graphics capability, memory and performance appropriate to specialist software.

Project 1 | Example hardware choices

Administrative Assistant

- AMD Ryzen 5 5500 CPU
- 16 GB DDR4 RAM; 1 TB NVMe SSD
- 27-inch MSI PRO display
- Keyboard, mouse and USB headset
- Recorded build total: £595.83*

Project Manager

- AMD Ryzen 7 5700 CPU
- 32 GB DDR4 RAM; 1 TB NVMe SSD
- GT 1030 graphics card
- Ethernet, webcam and headset
- Recorded total for 2 units: £1,707.82*

*Prices and totals are reproduced from the supplied project documents; verify component prices and arithmetic before procurement.

Project 1 | Specialist workstations and budget

- Architect/CAD Designer: documented hardware cost £374.51 per workstation; quantity 5; total £1,872.55.
- 3D Visualisation Specialist: £2,290.39 per workstation; quantity 2; total £4,580.78.
- Administrative Assistant: recommended quantity 3; team recommendation lists £589.84 each and £1,769.52 total.
- Project Manager: quantity 2; £853.91 each and £1,707.82 total.
- The team's final summary states a hardware refresh total of £9,930.67 against a £26,000 hardware-only budget.
- Note: the Administrative Assistant build list shows a different total (£595.83); reconcile this discrepancy before final purchasing.

Project 1 | SQL database and relationships

- Database records employee roles, workstation builds, build items and recommendations.
- Primary and foreign keys link roles to builds and builds to individual items.
- Queries retrieve all roles, role budgets, team sizes, build contents and item counts.
- Aggregate queries use SUM and GROUP BY to calculate build and role costs.
- JOIN queries combine role, build and item information for cross-table reporting.

Project 1 | What the SQL work demonstrates

Data retrieval and filtering

- List every role and its details.
- Show only role name and budget.
- Filter roles with budgets over £1,000 per workstation.
- Sort roles by team size.

Summaries and joins

- Find items for one build and count them.
- Calculate total cost per build.
- Join build names to their roles.
- Sum item costs by role using JOIN, SUM and GROUP BY.

Project 2 | Pennine Pizza Works ordering program

- Purpose: collect customer details, capture a pizza order and print an itemised bill.
- The brief supports orders of 1–7 pizzas, each with a menu choice, size and 1–7 distinct toppings.
- The bill includes customer information, pizza count, pizza sizes, toppings, item charges, subtotal, delivery fee and grand total.
- Currency is formatted to two decimal places; the solution does not process online payments.

Project 2 | Menu and pricing rules

Pizza size base prices

- Small — £10.00
- Medium — £15.00
- Large — £20.00
- Choose one size for each pizza.

Topping prices

- Sausage — £2.00; Chilli — £0.50
- Ham — £1.50; Pineapple — £3.30 in the analysis data
- Olives — £2.00; Pepperoni — £0.50; Tomatoes — £0.50
- The sample run lists pineapple as £3.50; confirm the intended price.

The source documents contain a pineapple price inconsistency (£3.30 vs £3.50), which should be clarified with the client.

Project 2 | Program design and validation

- Main coordinates the process: load menu data → get customer details → select pizzas → calculate costs → display bill.
- Reusable functions include CheckBlank, ReadInteger, ShowMenu, ChoosePizzas, CalculateCost, DisplayBill and Money.
- Input validation rejects blank customer fields, non-integers and choices outside allowed ranges.
- Repeated toppings on the same pizza are rejected to prevent accidental double charging.
- Prices are stored as integer pence to avoid floating-point rounding errors.

Project 2 | Cost calculation and assumptions

- Each pizza cost = size base price + selected topping prices.
- Order subtotal = sum of all individual pizza costs.
- Delivery fee is assumed to be £0.00 because the brief provides no delivery-charge rule.
- No taxes or discounts are applied because no rules were supplied.
- Invalid input prompts a helpful error message and asks the user to try again.
- Testing should cover minimum/maximum pizza counts, invalid choices, duplicate toppings and correct bill totals.

Project 3 | Network model and objectives

- Northwest Architecture Studio network model was built and tested in Filius.
- The model covers office LAN devices, servers, DNS, switches, a router, gateway and external network access.
- The aim is to understand addressing, name resolution and how devices communicate inside and outside the office.
- Ping/ICMP tests are used to check whether hosts respond and to observe connectivity or packet loss.

Project 3 | Key networking concepts

LAN equipment and routing

- A switch connects devices within the same LAN (Layer 2).
- A router forwards packets between networks using IP addresses (Layer 3).
- A gateway is the route a client uses to reach destinations beyond its local network.
- NAT maps private internal addresses to a public address for external communication.

Addressing and services

- An IP address identifies a network interface and helps deliver packets.
- DNS translates human-readable domain names into IP addresses.
- Client-server architecture: a client requests a resource; a server responds.
- HTTP and SQL are examples of application-layer protocols referenced in the project.

Project 3 | Connectivity tests

- The documented tests include LAN PC-to-PC, PC-to-office-server and PC-to-DNS-server.
- The model also tests internal web access by IP and by name.
- External server ping and external web access by IP and by name are included.
- Comparing IP-based and name-based tests helps distinguish basic reachability from DNS/name-resolution issues.
- The project includes an IP address table and a diagram of the Filius network.

Project 4 | Safe use of electronic devices

- Devices bring benefits for communication, learning, work and entertainment, but unsafe use can create electrical, fire, physical, ergonomic, hearing, privacy and cybersecurity risks.
- Inspect plugs, cables, chargers and sockets; stop using equipment with burn marks, sparks, unusual heat, smells or damaged cables.
- Use compatible chargers and charge batteries on a hard, flat surface away from bedding and water.
- Do not use swollen, leaking, smoking or abnormally hot batteries.
- Set up screens and workstations ergonomically, take short frequent breaks and protect hearing at high headphone volumes.

Project 4 | Authentication: MFA and 2FA

Authentication factors

- Something you know: password, passphrase or PIN.
- Something you have: device or physical security key.
- Something you are: fingerprint or facial recognition.
- 2FA uses two distinct authentication factors; MFA adds more than one factor.

Strengths and limitations

- MFA substantially reduces account takeover risk compared with passwords alone.
- Traditional SMS codes and OTPs can be vulnerable to phishing and SIM-swapping.
- Passkeys and FIDO2 security keys provide stronger phishing resistance by binding authentication to the legitimate website.
- Device protection and account recovery still matter.

Project 4 | Public Wi-Fi security

- Threats include man-in-the-middle attacks, fake “evil twin” hotspots, packet sniffing, phishing and malicious captive portals.
- Possible consequences: account compromise, identity theft, financial fraud and exposure of organisational data.
- Verify the network name with staff before connecting and check website addresses and HTTPS warnings.
- Keep operating systems, browsers and apps updated; disable automatic connection to unknown hotspots.
- A reputable VPN can add protection on untrusted networks but does not prevent phishing or malware.
- For sensitive tasks, consider mobile data or a personal hotspot instead.

Project 4 | Phishing awareness

- Phishing messages imitate trusted organisations to persuade people to disclose credentials, open harmful links or provide information.
- Check the sender and domain carefully; be suspicious of urgency, unexpected attachments and requests to bypass normal processes.
- Do not enter passwords after following an unexpected link; navigate to the official service independently.
- Report suspicious emails or messages through the organisation’s agreed process.
- The source project also covers related scams such as business email compromise and smishing (SMS phishing).

Combined recommendations

- Hardware: confirm component compatibility, current prices and role requirements before purchasing.
- Database: maintain accurate keys and relationships; use queries to support transparent cost reporting.
- Ordering software: confirm ambiguous business rules, especially delivery fees and the pineapple topping price, then test boundary and error cases.
- Networking: document IP addressing and verify DNS, internal and external connectivity separately.
- Security: combine safe device handling, software updates, MFA, verified Wi-Fi and user awareness training.

Conclusion

- The four projects connect practical IT decisions with software design and cybersecurity awareness.
- Project 1 plans role-appropriate workstations and uses SQL to report inventory and costs.
- Project 2 applies decomposition, validation, arrays and reusable functions to an ordering problem.
- Project 3 demonstrates LAN, routing, IP addressing, DNS and client-server communication.
- Project 4 focuses on safer device use, stronger authentication and secure public Wi-Fi habits.
- Together, they show the value of reliable design, clear assumptions, testing and responsible technology use.

References and source notes

- Primary source materials: Mini Project 1 & 3 markdown document; Team 1 Software Engineering Mini Project 2 PDF; MiniProject4 PDF.
- Project 4 cites guidance from HSE, NCSC, GOV.UK, Electrical Safety First, WHO, FTC, CISA and NHS England, along with research on MFA and account-takeover prevention.
- Figures and price values are based on the submitted project documents; several inconsistencies are flagged in this presentation rather than silently resolved.
- Add team member names, tutor, date and any required institutional branding before presenting.